Problem sheet 1 for Rigid Analytic Geometry Summer Term 2026

As in the first lecture we consider fields K with an absolute value $|\cdot|_K$ such that the following holds for $x, y \in K$:

- $|x+y|_K \leq \max(|x|_K, |y|_K)$
- $|xy|_K = |x|_K |y|_K$.
- $|0|_K = 0$ and $|1|_K = 1$.
- K becomes a complete and non-discrete metric space with with metric $d(x,y) = |x-y|_K$.

Let

$$K^\circ = \{x \in K \mid |x| \leq 1\}$$
$$K^{\circ\circ} = \{x \in K \mid |x| < 1\}$$

be the subring of power-bounded elements of K and $K^{\circ\circ}$ its ideal of topologically nilpotent elements.

Problem 1 (5 points)**.** *Show that* SpecK° *contains precisely the two elements* $\{0\}$ *and* $K^{\circ\circ}$.

In particular, K° is a local ring.

By a *K-Banach space* we understand a K-vector space E with a map $E \xrightarrow{\|\cdot\,|E\|} [0,\infty)_{\mathbb{R}}$ such that the following holds for $x, y \in E$ and $\lambda \in K$:

- $\|x+y\,|E\| \leq \max(\|x\,|E\|, \|y\,|E\|)$
- $\|\lambda x\,|E\| = |\lambda|_K \|x\,|E\|$.
- E becomes a complete metric space with with metric $d(x,y) = \|x-y\,|E\|$.

A subset X of a K-Banach space is called *bounded* if $\{\|x\,|E\| \mid x \in X\}$ is bounded subset of $\mathbb{R}$.

A *K-Banach algebra* is a K-algebra[1] A equipped with a K-Banach space norm $\|\cdot\,|A\|$ such that

$$\|ab\,|A\| \leq \|a\,|A\| \, \|b\,|A\| \tag{1}$$
$$\|1\,|A\| \leq 1 \tag{2}$$

holds for $a, b \in A$. The norm is called *multiplicative* if equality always occurs in (1) and (2). It is called *power multiplicative* if $\|a^n\,|A\| = \|a\,|A\|^n$ for all $a \in A$ and $n \in \mathbb{N}$.

[1] We will only be interested in the case where A is commutative and has a unit element.

Remark 1.
- *It easily follows from* (1) *and* (2) *that* $\|1\,|A\| = 1$ *or* $\|1\,|A\| = 0$ *where the second case will occur precisely if* $A = \{0\}$.
- *Let* N *be a* K*-Banach space norm on* A *such that*

$$N(ab) \leq CN(a)N(b)$$

for some $C \in (0,\infty)_{\mathbb{R}}$. *If* $A \neq \{0\}$ *one puts*

$$\|x\| = \sup_{y \in A\setminus\{0\}} \frac{N(xy)}{N(y)}$$

one gets a norm satisfying (1) *and* (2) *which is equivalent to* N *as*

$$N(x) \leq N(1)\,\|x\| \qquad\qquad \|x\| \leq CN(x).$$

Thus (1) *and* (2) *are essentially a "without losing generality" condition which we impose following the textbook "Nonarchimedean Analysis" by Bosch, Güntzer and Remmert which is the standard source for the foundations of classical rigid analytic geometry.*

Let

$$\mathbb{T}_m = \Big\{ f = \sum_{\alpha \in \mathbb{N}^m} f_\alpha X^\alpha \in K[\![X_1,\ldots,X_m]\!] \;\Big|\; \text{For all } \varepsilon \in (0,\infty)_{\mathbb{R}} \text{ there are only finitely many } \alpha \in \mathbb{N}^m \text{ such that } |f_\alpha| \geq \varepsilon. \Big\}.$$

Problem 2 (6 points)**.** $\mathbb{T}_m$ *is a* K*-Banach algebra and* $\|\cdot\,|\mathbb{T}_m\|$ *is multiplicative.*

Problem 3 (3 points)**.** *If* A *is a* K*-Banach algebra then its subset of power-bounded elements*

$$A^\circ = \Big\{ a \in A \;\Big|\; \{a^n \mid n \in \mathbb{N}\} \text{ is bounded in } A \Big\}$$

is a subring and its subset

$$A^{\circ\circ} = \Big\{ a \in A \;\Big|\; \lim_{k\to\infty} a^k = 0 \Big\}$$

of topologically nilpotent elements is an ideal in A° *satisfying* $\sqrt{A^{\circ\circ}} = A^{\circ\circ}$.

Problem 4 (1 point)**.** *If* A *is a* K*-Banach algebra with a power multiplicative norm then*

$$A^\circ = \big\{ a \in A \;\big|\; \|a\,|A\| \leq 1 \big\}$$
$$A^{\circ\circ} = \big\{ a \in A \;\big|\; \|a\,|A\| < 1 \big\}.$$

Problem 5 (2 points)**.** *A* K*-Banach algebra has no proper open ideal.*

Problem 6 (3 points)**.** *The group of units of a K-Banach algebra is a neighbourhood of* 1*.*

Problem 7 (2 points)**.** *A K-Banach algebra has no proper dense ideal.*

In particular, if follows from the above that $\mathbb{T}_m$ has no proper open ideal (as was claimed in the lecture) and that it has no proper dense ideal.

Two of the 22 points from this sheet are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday April 27.